Deep Learning for Orientation Mapping in 4D-STEM

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1 Problem Understanding

A 4D-STEM experiment can generate tens of thousands of diffraction images, each containing information about the orientation of a tiny crystal grain. Our goal is to train a convolutional neural network (CNN) to infer the three Bunge Euler angles (ϕ_1, Φ, ϕ_2) directly from each image quickly and accurately, without relying on computationally expensive template [3]. This is challenging because of angle periodicity, crystal symmetry ambiguity, the sparsity of the diffraction signal and the massive scale of the dataset 1. A robust solution would greatly accelerate orientation mapping and could enable real-time materials characterisation.

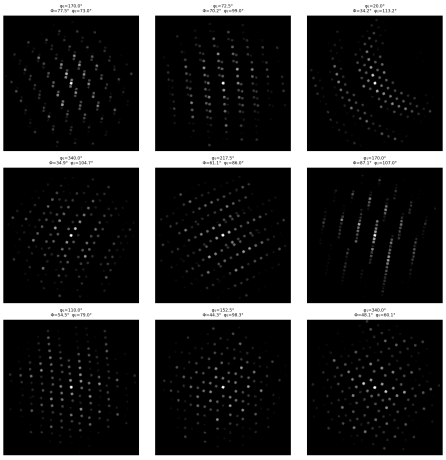

Fig. 1: Sample diffraction patterns with corresponding Euler angles from dataset.

2 Data Exploration

We explored the dataset before any modelling to understand its structure, label distribution, and inherent challenges.

2.1 Dataset Overview

The dataset [1] contains 581,328 simulated $LiNiO_2$ diffraction patterns. Each image is a 256×256 grayscale PNG file. Ground truth Euler angles are encoded directly in the filename and parsed using regular expressions. Table 1 summarises the key properties.

Table 1: Dataset properties of the $LiNiO_2$ diffraction dataset.

Property	Value
Total images	581,328
Image size	256×256 pixels (grayscale)
Format	PNG, `uint8`
Label source	Parsed from filenames
Thickness	70 nm (fixed)
ϕ_1 **range**	$[0°, 360°)$, step $2.5°$
Φ **range**	$[0°, 90°]$
ϕ_2 **range**	$[60°, 120°]$
Near-zero pixels ($< 5/255$)	$\approx 94\%$
Pixel mean	≈ 2.8
Pixel std	≈ 14.6

2.2 Orientation Distribution

Figure 2 shows the marginal distributions of the three Euler angles. ϕ_1 is perfectly uniform across $[0°, 360°)$ by design. Φ increases toward $90°$, reflecting the solid-angle geometry of orientation space. ϕ_2 is restricted to $[60°, 120°]$, covering only the fundamental zone of $LiNiO_2$'s hexagonal crystal symmetry.

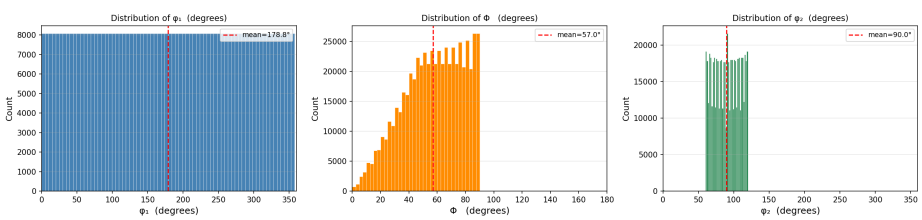

Fig. 2: Euler angle distributions. ϕ_1 is uniform by design. Φ rises toward $90°$ (solid-angle geometry). ϕ_2 is restricted to $[60°, 120°]$ by crystal symmetry.

2.3 Joint Orientation Coverage

Figure 3 shows the pairwise joint distributions of the three Euler angles. The ϕ_1–Φ and ϕ_1–ϕ_2 plots show uniform rectangular coverage, confirming that ϕ_1

is independent of both other angles by design. The Φ–ϕ_2 plot, however, shows a non-uniform structure: at low Φ (near 0°) very few ϕ_2 values exist, while at high Φ (near 90°) the full ϕ_2 range is densely populated. This confirms that Φ and ϕ_2 are coupled by crystal symmetry and cannot be treated as independent regression targets.

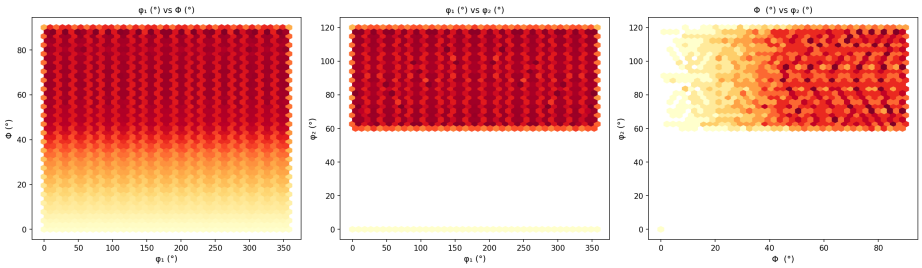

Fig. 3: Joint distributions of Euler angle pairs across all 581,328 images.

3 Methodology

Our approach frames orientation prediction as a regression problem where a CNN maps a single diffraction image directly to three Euler angles. To handle the periodicity of angles, each angle is encoded as a $(\sin, \cos)$ pair before training and decoded back using atan2 after prediction. Figure 4 shows the complete end-to-end pipeline.

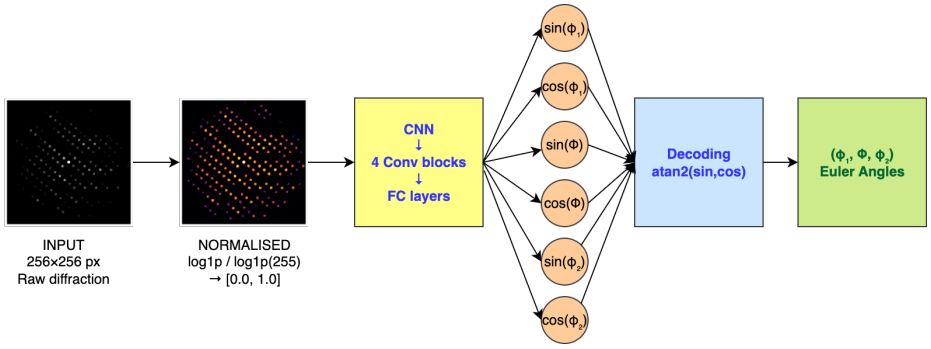

Fig. 4: End-to-end pipeline

3.1 Problem Formulation

Orientation prediction is a regression problem over three continuous Euler angles. Table 2 compares the three candidate formulations considered.

We choose **sin/cos regression** because orientation is continuous, raw angle regression is broken by periodicity, and classification requires an impractically large number of classes.

Table 2: Comparison of problem formulations for orientation prediction.

Approach	Advantages	Disadvantages
Classification [2]	Simple formulation; symmetry can be handled by merging equivalent classes	Extremely large label space ($>$ 580,000 classes); no ordinal structure; cannot interpolate between classes
Raw regression	Direct prediction with only three outputs	Fails at periodic boundaries ($0°/360°$); unstable gradients
Sin/cos regression (ours)	Smooth and bounded representation; handles periodicity naturally; stable gradients	Requires six outputs instead of three; decoding step needed

3.2 Label Representation

Sin/cos encoding. Each Euler angle ϕ is encoded as a $(\sin\phi,\ \cos\phi)$ pair, yielding six continuous outputs per image:

$$\hat{\mathbf{y}} = \left[\sin\phi_1,\ \cos\phi_1,\ \sin\Phi,\ \cos\Phi,\ \sin\phi_2,\ \cos\phi_2\right] \in [-1,+1]^6 \tag{1}$$

This representation avoids discontinuities at $0°/360°$ and provides a smooth, bounded target space for regression [5].

Decoding. After prediction, angles are recovered exactly using:

$$\hat{\phi} = \operatorname{atan2}(\hat{s},\ \hat{c}) \times \frac{180°}{\pi} \bmod 360° \tag{2}$$

Experiment 3 modification. In Experiment 3, ϕ_1 is reduced to the fundamental segment before encoding:

$$\phi_1^{\text{red}} = \phi_1 \bmod 60° \quad \Rightarrow \quad \phi_1^{\text{red}} \in [0°,\ 60°) \tag{3}$$

This maps all six symmetry-equivalent ϕ_1 values to the same label, removing the 6-fold ambiguity before training.

3.3 Data Preparation

Log-scaling. As identified in data exploration, $\approx 94\%$ of pixels have intensity below 5 out of 255.

We apply a logarithmic transformation:

$$x' = \frac{\ln(1+x)}{\ln(256)} \in [0.0,\ 1.0] \tag{4}$$

This redistributes pixel values more evenly and makes faint spots visible to the convolutional filters. Figure 5 shows the effect on a representative image.

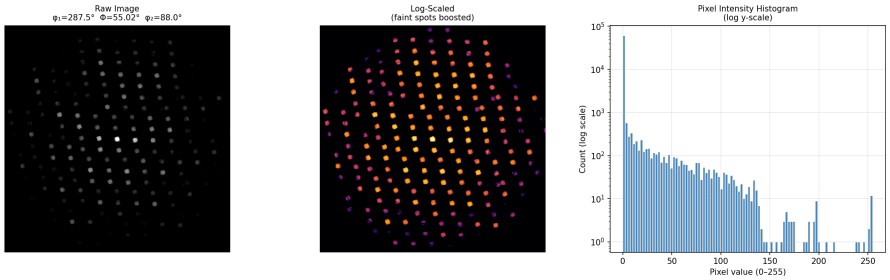

Fig. 5: Effect of log-scaling on a single diffraction image.

4 Model Design

4.1 Data Splits

To ensure fair comparison across experiments, we use consistent train, validation, and test splits. Each dataset is randomly partitioned into 80% training, 10% validation, and 10% test samples. Table 3 summarises the exact splits used in each experiment.

Table 3: Data splits across experiments.

Experiment	Total	Train	Validation	Test
Experiment 1	10,000	8,000	1,000	1,000
Experiment 2	30,000	24,000	3,000	3,000
Experiment 3	10,000	8,000	1,000	1,000

4.2 Architecture

We use a custom convolutional neural network (OrientationCNN) to map diffraction images to Euler angles. CNNs are well suited for this task as they capture local features while remaining translation-equivariant.

The network consists of four convolutional blocks followed by global average pooling and a fully connected regression head, as summarised in Table 4.

4.3 Training Strategy

Loss function. Experiments 1 and 3 use Mean Squared Error over all six sin/cos outputs:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{6}\sum_{i=1}^{6}(\hat{y}_i - y_i)^2 \tag{5}$$

MSE on $(\sin\phi, \cos\phi)$ is geometrically equivalent to the squared chord distance between two points on the unit circle, which is a monotone function of the angular distance.

Table 4: Architecture of OrientationCNN.

Layer	Output shape	Description
Input	$(B, 1, 256, 256)$	Log-scaled grayscale image
ConvBlock 1	$(B, 32, 128, 128)$	2×(Conv-BN-ReLU), MaxPool
ConvBlock 2	$(B, 64, 64, 64)$	2×(Conv-BN-ReLU), MaxPool
ConvBlock 3	$(B, 128, 32, 32)$	2×(Conv-BN-ReLU), MaxPool
ConvBlock 4	$(B, 256, 16, 16)$	2×(Conv-BN-ReLU), MaxPool
GAP	$(B, 256)$	Global average pooling
FC 1	$(B, 512)$	Linear + ReLU + Dropout ($p = 0.3$)
FC 2	$(B, 256)$	Linear + ReLU + Dropout ($p = 0.3$)
Output	$(B, 6)$	Linear (sin/cos outputs)

Total parameters: ≈ 1.2M

Experiment 2 replaces this with a **symmetry-aware loss** that accepts any of the six equivalent ϕ_1 values as equally correct:

$$\mathcal{L}_{\text{sym}}(\hat{\phi}_1, \phi_1) = \min_{k \in \{0,1,2,3,4,5\}} \mathcal{L}_{\text{MSE}}\Big(\hat{\phi}_1,\ (\phi_1 + k \cdot 60^\circ) \bmod 360^\circ\Big) \tag{6}$$

5 Results

5.1 Training and Validation

Figure 6 shows the training and validation loss curves for all three experiments.

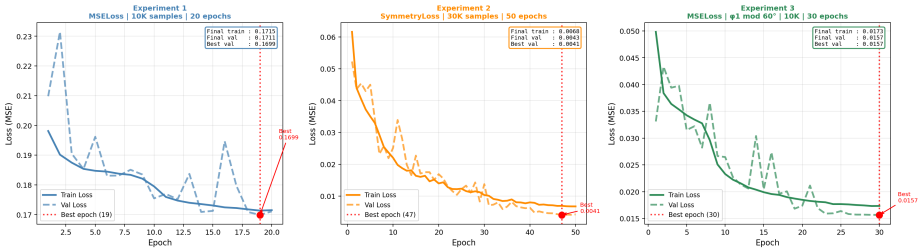

Fig. 6: Training loss (solid) and validation loss (dashed)

Three observations are consistent across all experiments. Training and validation losses remain closely aligned, indicating no overfitting and effective regularisation. Validation loss is noisier, which is expected due to the small validation set and task difficulty. Finally, Experiment 2 achieves a validation loss of 0.0041, over 40× lower than Experiment 1 (0.1699), showing that increased data and training improve convergence.

5.2 Experiment Evaluation on Test Set

Table 5 reports the mean angular error (MAE) for each Euler angle on the held-out test set across all three experiments, compared against a random guessing baseline.

Table 5: Test set performance across experiments.

Metric	Exp 1	Exp 2	Exp 3
ϕ_1 **MAE (°)**	88.60	88.52	14.83
Φ **MAE (°)**	1.75	1.28	0.77
ϕ_2 **MAE (°)**	4.82	2.58	2.14
Overall MAE (°)	31.72	30.79	5.91

Figure 7 shows the predicted versus true values for each Euler angle across all three experiments on the test set. Each row corresponds to one experiment and each column to one angle. A perfect model would place all points exactly on the dashed diagonal line.

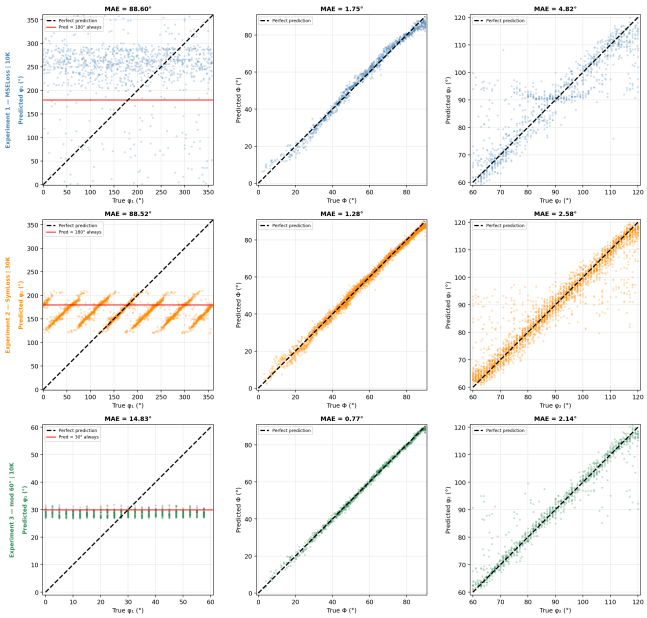

Fig. 7: Predicted vs. true Euler angles on the test set.

5.3 Discussion and Insights

Φ and ϕ_2 are learnable. Both angles improved consistently across all three experiments. In Experiment 3, the model achieved an MAE of 0.77° for Φ and 2.14°

for ϕ_2. These results confirm that the CNN architecture and sin/cos encoding are well suited for predicting these two angles.

ϕ_1 is not learnable. ϕ_1 remained at the random baseline in every experiment. This behaviour did not change with different loss functions, data sizes, or output space configurations. The reason is a fundamental physical limitation. $LiNiO_2$ has hexagonal symmetry [4], which means that rotating ϕ_1 by 60° produces an identical diffraction pattern. Figure 8 provides direct visual proof. Six patterns with different ϕ_1 values appear identical when $\Phi \approx 90°$.

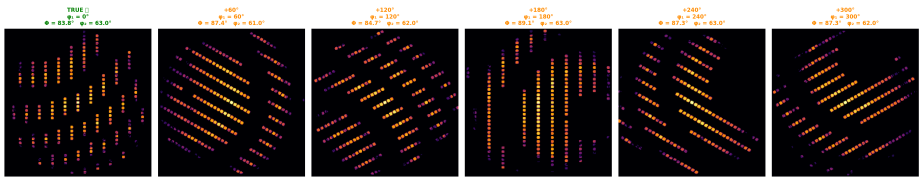

Fig. 8: Six symmetry-equivalent diffraction patterns.

The failure mode is detectable. In every worst case, the model predicted $\phi_1 \approx 180°$ in Experiments 1 and 2, and $\approx 30°$ in Experiment 3. Both values correspond to the centre of the respective output range. This mean-prediction behaviour is a reliable signal of uncertainty. It could be used in practice to automatically flag unreliable scan points.

6 Conclusion

We trained a convolutional neural network to predict crystal orientation directly from $LiNiO_2$ diffraction patterns. Sin/cos encoding was used to handle the periodicity of Euler angles. Three experiments were conducted to progressively address the challenges of symmetry and output space design.

Φ and ϕ_2 were predicted reliably across all experiments. The best results were an MAE of 0.77° for Φ and 2.14° for ϕ_2. ϕ_1 could not be learned in any experiment. This is because $LiNiO_2$'s hexagonal symmetry produces identical diffraction patterns for six different ϕ_1 values. This is a physical limitation of the data, not a limitation of the model.

The results show that deep learning can effectively replace template matching for two out of three orientation angles. They also highlight that understanding the physical symmetry of the material is essential for designing a successful approach.

References

1. J. Scheunert, *CNNs for Orientation Mapping of LNO Electron Diffraction Patterns Together with Test-data*, Zenodo, Oct. 2025. DOI: 10.5281/zenodo.17360572
2. J. Scheunert, S. Ahmed, T. Demuth, A. Beyer, S. Wissel, B.-X. Xu, and K. Volz, *Determining the grain orientations of battery materials from electron diffraction patterns using convolutional neural networks*, *npj Computational Materials*, 2026.
3. E. F. Rauch and M. Véron, *Automated crystal orientation and phase mapping in TEM*, *Materials Characterization*, vol. 98, pp. 1–9, 2014. DOI: 10.1016/j.matchar.2014.08.010
4. A. Rougier, P. Gravereau, and C. Delmas, *Optimization of the Composition of the $Li_{1-z}Ni_{1+z}O_2$ Electrode Materials: Structural, Magnetic, and Electrochemical Studies*, *Journal of The Electrochemical Society*, vol. 143, no. 4, pp. 1168–1175, 1996. DOI: 10.1149/1.1836614
5. Y. Zhou, C. Barnes, J. Lu, J. Yang, and L. Li, *On the Continuity of Rotation Representations in Neural Networks*, in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5745–5753, 2019. DOI: 10.1109/CVPR.2019.00589